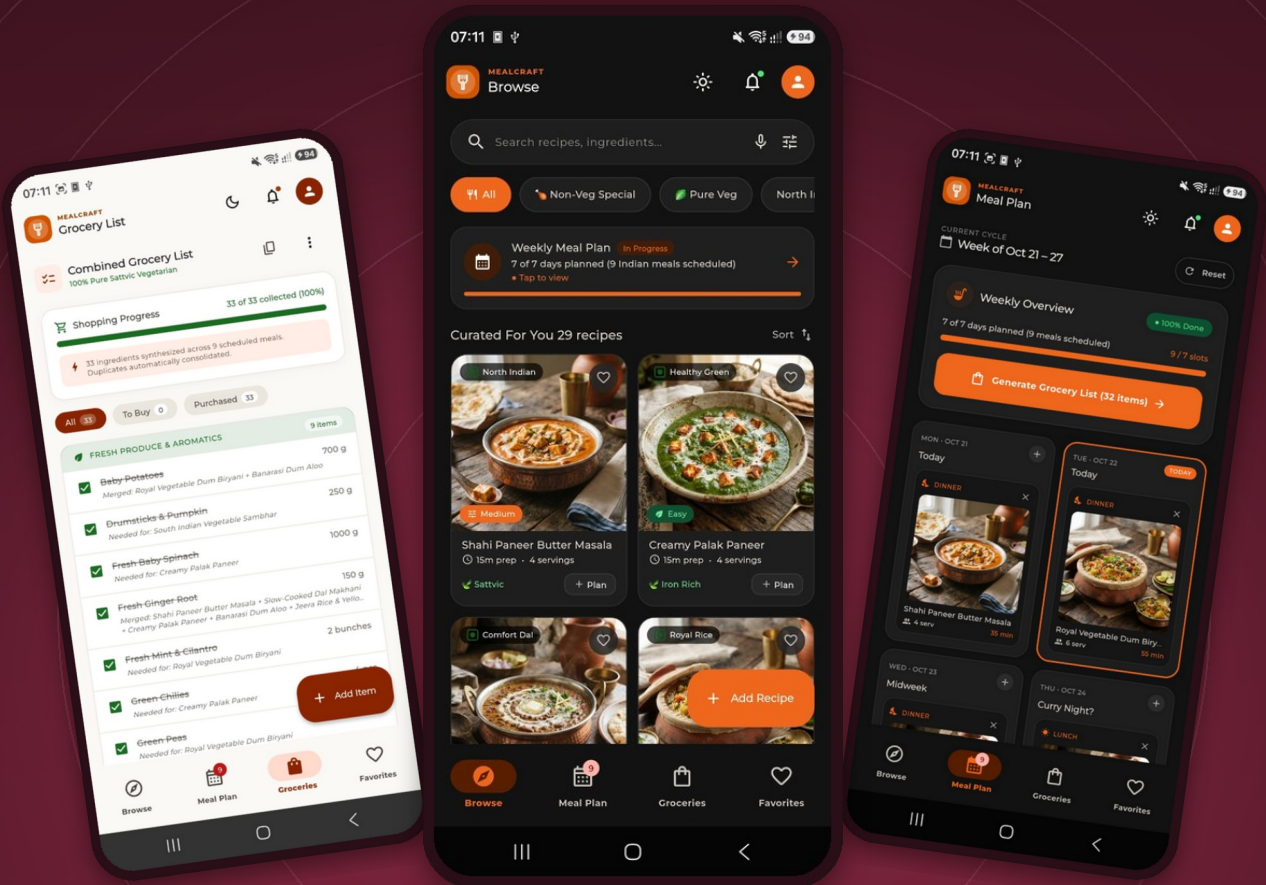


MealCraft

Plan the week. Merge the groceries. Cook with confidence.

PROBLEM STATEMENT 40 • SUBMISSION



29

Indian recipes

3

platforms, 1 codebase

$O(N)$

grocery merge

Rohan Vashisht

Roll No.: 150096724132 • Cohort: Jensen Huang

B.Tech CSE, Semester V • 2024.rohanv@isu.ac.in



LIVE APP

rohan-cross-platform.vercel.app



GITHUB REPOSITORY

[rohanvashisht1234/cross_platform_app_development](https://github.com/rohanvashisht1234/cross_platform_app_development)

Problem Statement and Motivation

1

Planning chaos

Busy urban households decide meals day by day, with no single weekly view of what to cook and when.

2

Ingredient wastage

The same ingredient appears in many recipes, so items are bought twice, in excess, or forgotten at the supermarket.

3

No automated grocery synthesis

Recipe apps list ingredients per recipe; shoppers add up quantities by hand and servings rarely scale.

What PS-40 demands

- Cross-platform recipe sharing
- A weekly meal planner
- Grocery consolidation: duplicate ingredients merged across all planned recipes
- One app for Android, iOS and Web

Solution Overview and Architecture

Presentation

Flutter UI, Material 3: Browse, Meal Plan, Groceries, Favorites, Profile



State logic

In-memory reactive state with offline persistence hooks



Services

IngredientMerger and SampleData



Data models

Recipe, Ingredient, MealPlan, MergedIngredient

Figure 1: Layered architecture of MealCraft

● One codebase

A single Flutter 3 / Dart project builds Android, iOS and Web.

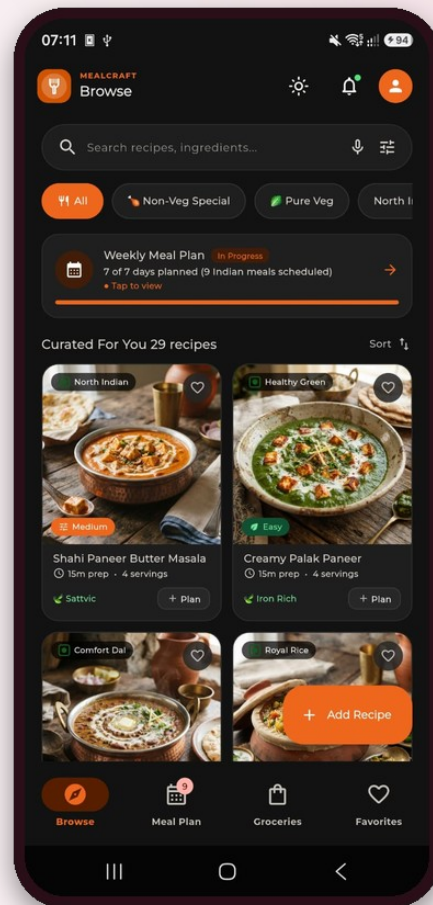
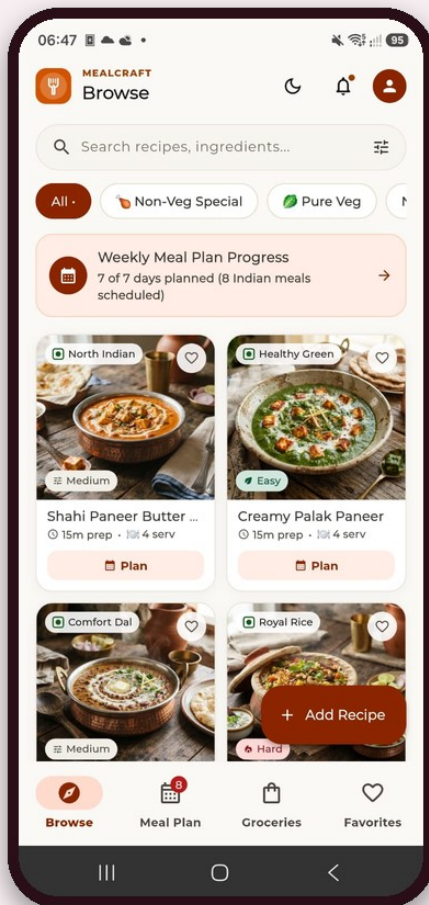
● Separation of concerns

The UI never merges ingredients itself; it calls the service layer.

● Testable core

IngredientMerger is a pure function, so it is testable without any UI.

Recipe Browsing and Filtering



1

2-column GridView

Recipe cards with photo, cuisine badge, difficulty tag, prep time and servings.

2

Filter chips

All, Non-Veg Special, Pure Veg and cuisines such as North Indian, Biryani and Rice, Street Food.

3

Live search

Search by recipe or ingredient, with FSSAI-style green Veg and red Non-Veg marks.

4

One-tap planning

A Plan button on every card and a weekly progress banner: 7 of 7 days planned.

7-Day Weekly Meal Planner

MON

TUE

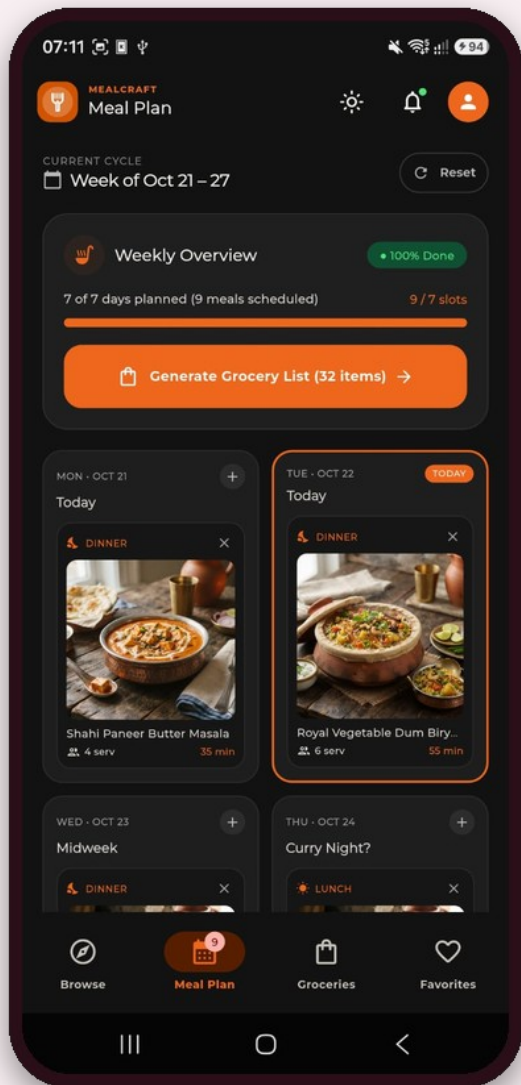
WED

THU

FRI

SAT

SUN



Uniform 2-column grid

Seven day cards, Monday to Sunday, with a TODAY highlight.

Three meal slots

Breakfast, Lunch and Dinner per day; add with plus, remove with a cross.

Weekly Overview

Progress bar: 7 of 7 days planned, 9 meals scheduled.

Week cycle

Cycle label such as Week of Oct 21 to 27, with a Reset button.

Generate Grocery List (32 items)

Planned meals flow straight into the IngredientMerger.

Map-Based Ingredient Merger

1

Normalize the key

Each ingredient name is lower-cased and trimmed, so Paneer and paneer become one key.

2

Merge or insert

A map keyed by that name is checked. If present, amounts are added and the recipe is listed once; if absent, a new entry is created.

3

Sort for display

Merged entries are sorted alphabetically for the shopping list.

Entry	Recipe	Amount
Paneer	Shahi Paneer (Monday lunch)	250 g
Paneer	Kadai Paneer (Friday dinner)	500 g
Merged: Paneer (Dairy)	Sources: Shahi Paneer, Kadai Paneer	750 g

Worked example: duplicates merge across the week

Complexity

Time $O(N)$: one pass over N ingredients with $O(1)$ average lookup. Space $O(U)$: one entry per unique ingredient.

Recipe Detail and Live Simmer Timer

Servings scaler



Tap 2, 4 or 8 servings. Every ingredient amount is rescaled instantly.

250 g paneer at 4 servings becomes 500 g at 8 servings.

Checkable steps



Prep the ingredients



Temper the spices



Simmer the gravy



Garnish and serve

Simmer timer



A built-in step timer counts down from 10:00 and raises an alert when the simmer is done.

Scaling formula

$\text{ScaledAmount} = \text{BaseAmount} \times (\text{SelectedServings} \div \text{BaseServings})$

Smart Supermarket Grocery Checklist

Produce

Dairy

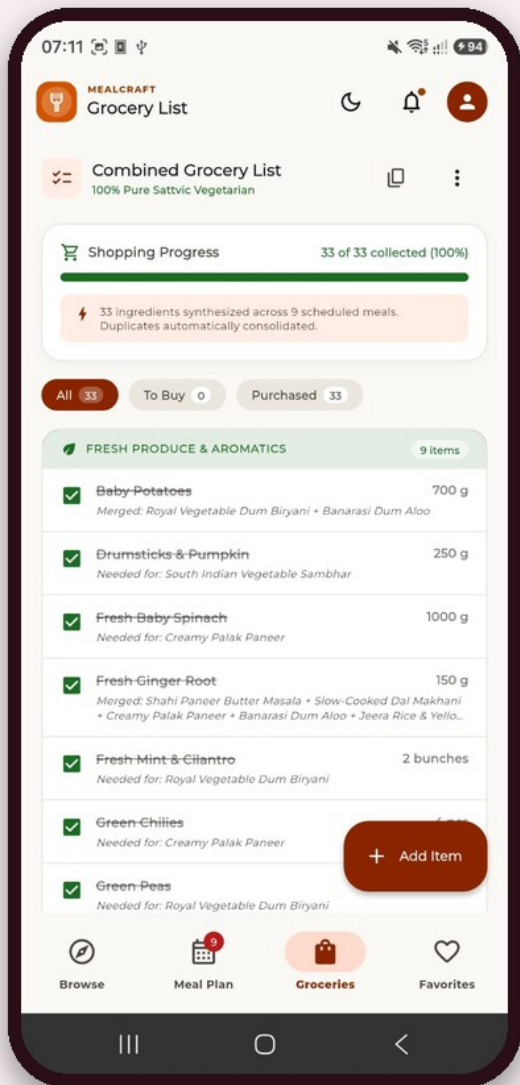
Spices

Grains

Proteins

Pantry

Other



● Aisle categories

Items grouped by aisle, such as Fresh Produce and Aromatics, with item counts.

● Shopping progress

33 of 33 collected (100%), with All, To Buy and Purchased tabs.

● Custom items

An Add Item dialog adds anything not found in a recipe.

● 1-tap export

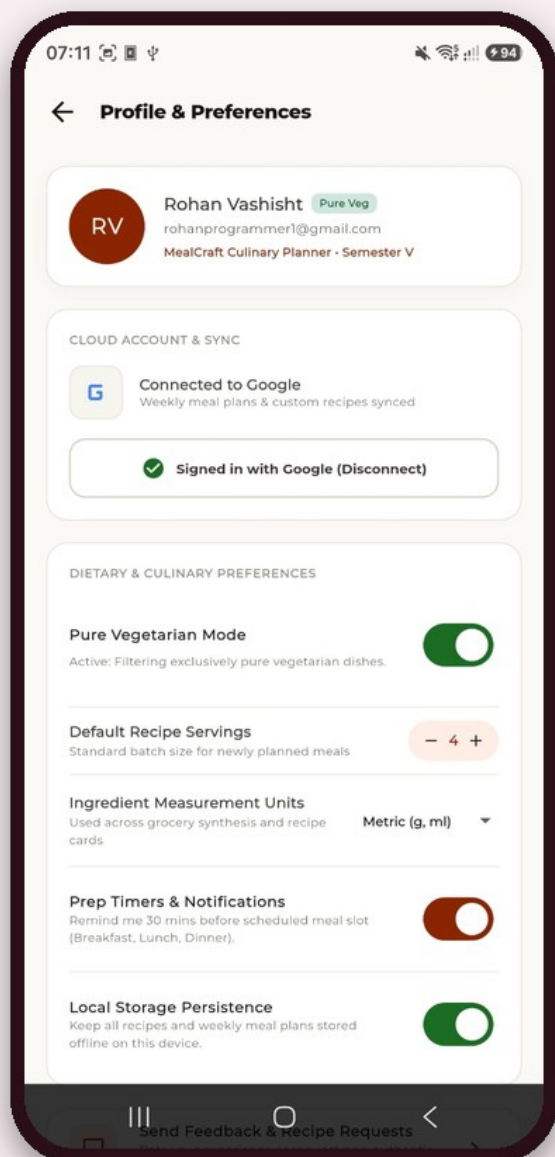
Copies the checklist as clean text for WhatsApp and Notes.

Merged badges show every source recipe

Duplicates are consolidated automatically across all 9 scheduled meals.

Settings, Google Auth and Theming

Design token	Light: Warm Terracotta	Dark: AMOLED
Primary	#8B2500 Terracotta	#EE671C Orange
Background	#FBF9F5 Cream	#121212 True Black
Surface	#FFFFFF White	#1E1E1E Charcoal
Success	#1B6D24 Green	#50E380 Mint



● Google Sign-In

Signed-in state shows weekly plans and custom recipes as synced.

● Pure Vegetarian mode

One toggle filters the app to pure vegetarian dishes only.

● Offline persistence

Local Storage toggle keeps recipes and plans on the device.

● Feedback dialog

Interactive 5-star rating and recipe request modal.

Cross-Platform Deployment on Vercel

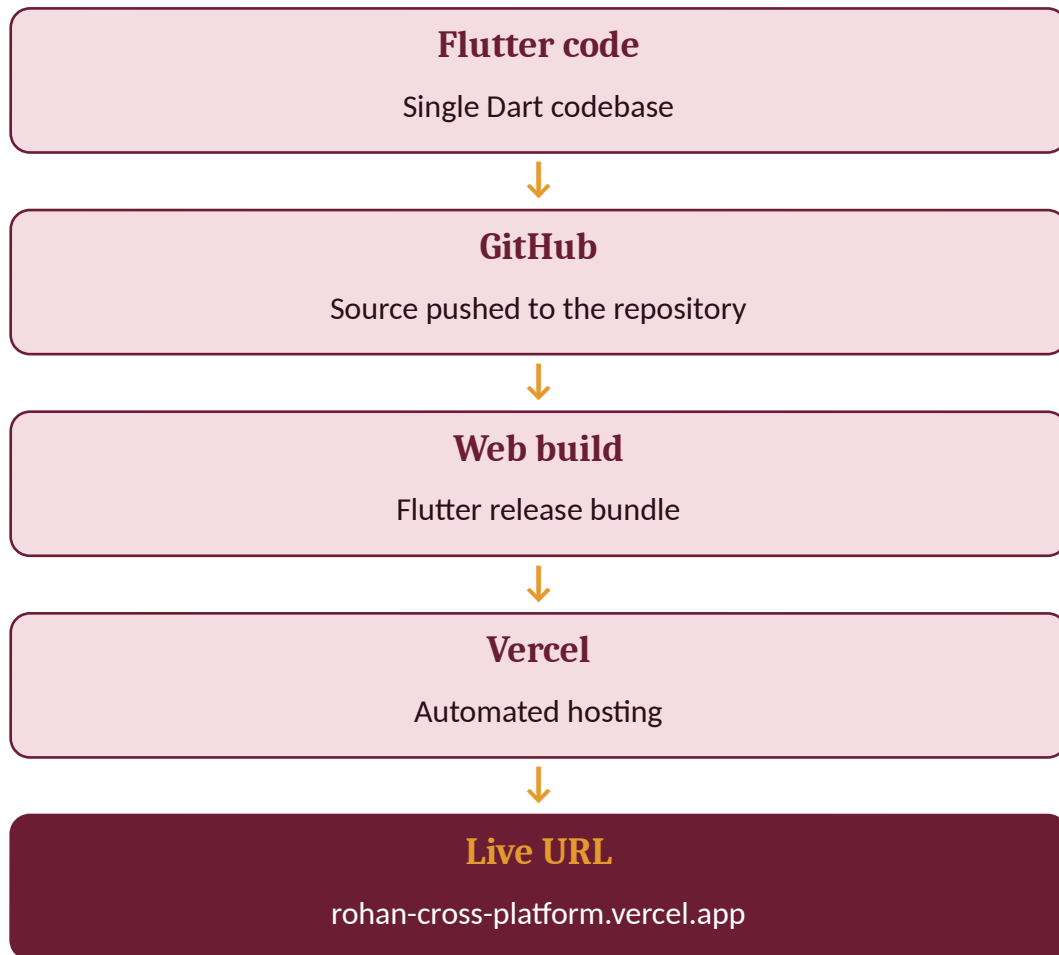


Figure 2: Git to live URL deployment pipeline

● Mobile

320 to 390 px screens: 2-column grid, bottom navigation, overflow-safe text.

● Tablet

Same Material 3 components; the grid adapts to the available width.

● Desktop

Runs in Chrome, Safari and Firefox from the Vercel URL.

Viva Questions and Answers

Q1

How does ingredient merging work with a Map?

Each name is normalized and used as the key. A hit adds the amount and lists the recipe once; a miss inserts a new entry. One pass over the ingredients gives $O(N)$ time.

Q2

StatefulWidget vs StatelessWidget here?

Stateless widgets draw fixed UI such as badges and cards. Stateful widgets hold changing data such as servings, timer, checklist ticks and theme, and rebuild when it changes.

Q3

How are RenderFlex overflows handled?

Long text is wrapped in flexible widgets with ellipsis, content is made scrollable, and layouts are tested at 320 to 390 px widths.

Q4

How is responsive design achieved?

Layout adapts to the screen width, so grid columns and spacing change while one codebase serves mobile and web.

Conclusion and Future Scope

What was achieved

- ✓ One Flutter codebase on Android, iOS and Web, live on Vercel
- ✓ Map-based merger consolidates a whole week in $O(N)$
- ✓ Servings scaler, step checklist and simmer timer
- ✓ Material 3 light and dark themes, 29 Indian recipes

● AI recipe generator

Suggest recipes from pantry items and dietary preferences.

● Supermarket API integration

Send the consolidated list to a grocery store for ordering.

● OCR receipt scanner and cloud auth

Scan receipts and add Supabase cloud sign-in and sync.

Thank you

[Live app: rohan-cross-platform.vercel.app](https://rohan-cross-platform.vercel.app)

[GitHub: rohanvashisht1234/cross_platform_app_development](https://github.com/rohanvashisht1234/cross_platform_app_development)